

Shageen Fathima S

Full-Stack Engineer | Backend-focused | Python/FastAPI | Node.js/NestJS | Microservices

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Full-Stack Engineer with 2+ years building production backend systems across fintech, HR-tech, and enterprise platforms. Delivered microservices handling financial transactions, bulk data processing, and **multi-tenant** workflows using **Node.js**, **NestJS**, **FastAPI**, **PostgreSQL**, **Kafka**, and **AWS**.

EDUCATION

Government College of Technology, Coimbatore

Bachelor of Technology — CGPA: 8.24 / 10

WORK EXPERIENCE

Full Stack Web Developer — KrishTech Software Solutions

May 2024 – May 2026

- Collaborated with designers and stakeholders to build responsive **Next.js** / **React.js** interfaces using **Tailwind CSS**, while handling frontend, backend, and database development end-to-end.
- Designed and developed REST APIs and microservices using **Node.js**, **NestJS**, and **PostgreSQL**, with **MongoDB** for NoSQL requirements and relational database schema design.
- Implemented **Apache Kafka** for asynchronous inter-service communication and **Redis** caching, reducing repetitive database operations by **~40%** and improving average API response time by **~30%**.
- Containerized applications using **Docker**, configured **Nginx** reverse proxy, and managed **AWS**-based deployment and cloud infrastructure end-to-end.
- Built payment gateway and subscription workflows using **Razorpay** and **Stripe**, alongside third-party integrations including **GST Fetch (IRIS)**, **Google OAuth**, **Twilio**, and **Puppeteer**.
- Developed PDF parsing and document-processing workflows using **OpenAI** to process extracted content and generate structured **CSV/Excel** documents.
- Implemented **JWT** authentication, middleware, role-based access control (**RBAC**), and protected routes for secure application access.

PROJECTS

All projects below were delivered as client engagements through KrishTech Software Solutions.

Amanah — Masjid Management Platform | Full-Stack Engineer — amanah.jamylabs.in

Tech Stack: Python, FastAPI, SQLAlchemy, PostgreSQL, Node.js, NestJS, GraphQL, Prisma, React.js, Next.js, Docker, AWS, Vercel, REST APIs, Git

- Developed and migrated a full-stack Masjid management platform covering Family/Member Management, Committees, **Sandha Payments**, Finance, Audit Logging, and Administrative Dashboards.
- Designed REST APIs using Python/**FastAPI** with **Controller → Service → Repository** architecture, backed by **PostgreSQL** and **SQLAlchemy**.
- Implemented transactional financial workflows connecting Sandha payments with the central **Transaction Ledger**, ensuring data consistency and integrity.
- Built complex APIs supporting relational queries, filtering, search, cursor pagination, aggregation, cascading operations, and audit tracking.

- Migrated existing **Next.js** / **Prisma** / **GraphQL** functionality to **FastAPI** / **SQLAlchemy** while preserving legacy business rules, validations, relationships, and audit requirements.

TeamLease — Background Verification (BGV) Platform | Full-Stack Engineer

Tech Stack: Node.js, NestJS, TypeScript, PostgreSQL, TypeORM, Redis, Kafka, GraphQL, React.js, Next.js, Keycloak, AWS, Git

- Developed **multi-tenant** BGV microservices using **NestJS**, **TypeScript**, **PostgreSQL**, and **TypeORM**, supporting enterprise onboarding and verification workflows across Business, Corporation, and Client hierarchies — utilized by **TeamLease and its enterprise clients across multiple business tenants**.
- Designed REST APIs and distributed service workflows, using **Apache Kafka** for inter-service communication across multiple microservices.
- Implemented **Redis**-based bulk processing for high-volume BGV uploads (**5,000+ records per batch**), including validation, processing, and failure handling — cutting failed-upload retries by **~20%** with downloadable failure reports.
- Implemented **Keycloak**-based authentication and role-based authorization, securing APIs across different operational roles.
- Worked on hierarchical configuration, master-switch inheritance, maker-checker approval workflows, and auditability across tenant and client levels.

Reveal — RFID Scanner-Based Production Management Application | Full-Stack Engineer — reveal.krishub.in

Tech Stack: Next.js, Tailwind CSS, Node.js, MongoDB, Flutter, Bluetooth Integration

- Designed and implemented **Node.js** REST APIs for shipment management, user management, production tracking, **RFID** tracking, and scan logging.
- Implemented role-based access control (**RBAC**) and protected routes for Shipper, Consignee, Agent, and Admin workflows.
- Designed scalable database schemas and optimized **MongoDB** operations using aggregation pipelines, indexing, and transactions.
- Integrated **RFID** scanners with **Flutter** and **Bluetooth**, enabling real-time **RFID** tag scanning and mapping to corresponding shipment boxes.
- Developed real-time operational workflows for tracking shipment items, **RFID** counts, and production status, improving visibility and data consistency.

SKILLS

- **Languages:** JavaScript, **TypeScript**, Python, PHP, HTML5, CSS3/SCSS
- **Frontend:** React.js, Next.js, React Native, Tailwind CSS
- **Backend & APIs:** Node.js, NestJS, FastAPI, REST APIs, GraphQL, API Integration
- **Architecture & Distributed Systems:** Microservices, **Apache Kafka**, **Redis**, Event-Driven Architecture, Asynchronous Workflows, TCP/IP
- **Databases & ORM:** PostgreSQL, MongoDB, Prisma, SQLAlchemy, TypeORM, Relational Data Modeling, Transactions, Complex Queries, Pagination & Filtering
- **Authentication & Security:** JWT, Keycloak, OAuth 2.0, Role-Based Access Control (**RBAC**), Protected Routes
- **Cloud, DevOps & Deployment:** Docker, AWS, Vercel, Nginx, Reverse Proxy, AWS CodeCommit
- **Payments & Integrations:** Razorpay, Stripe, Twilio, Google OAuth, IRIS GST Integration, Puppeteer, OpenAI
- **Tools & Version Control:** Git, GitHub, VS Code

CERTIFICATIONS

- KGISL — Software Trainee / Technical Trainer (MERN Stack): Conducted a 10-day MERN Stack training program covering **MongoDB**, Express.js, **React.js**, **Node.js**, REST APIs, and database integration.
- Skyappz Academy — Full Stack Web Development Intern: Practical exposure to **React.js** and **MongoDB**.
- NCC — B & C Certificates: Leadership, discipline, teamwork, and organizational training.

LANGUAGES

Tamil — Superior | English — Superior | Hindi — Intermediate